

Unlike solid rockets, there is no increase in p_c in the combustion chamber.

Recall: $\bar{M}_m = \sum X_i M_i$, $\bar{c}_{p,m} = \sum X_i \bar{c}_{p,i}$ $\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - \bar{R}}$

$$R_m = \frac{\bar{R}}{\bar{M}_m}$$

Example: An ideal H_2-O_2 rocket engine has $t = o/f = 5$.
Propellants are injected at 68 atm, and 298.15 K.

Use table 12.2 of textbook to read X_i & T_c of products

Assuming fully-expanded jet at sea level ($p_e = p_a = 1 \text{ atm}$)

Calculate I_{sp} . $\gamma_{mon} = 1.67$, $\gamma_{di} = 1.29$, $\gamma_{tri} = 1.17$

Solution: $p_o = 68 \text{ atm} = 68 \times 0.101325 \text{ MPa} = 6.89 \text{ MPa}$

From table 12.2 $T_c = 3429 \text{ K}$

" " " $X_{H_2} = 0.3607$
 $X_{O_2} = 0.977 \times 10^{-3}$
 $X_{H_2O} = 0.5744$
 $X_{OH} = 0.0260$
 $X_O = 0.015$
 $X_H = 0.0363$

CEA
 $T_c = 3423 \text{ K}$

Read these
 from CEA

$$\bar{M}_m = \sum X_i \bar{M}_i = 0.3607 \times 2 + 0.977 \times 10^{-3} \times 32 + 0.5744 \times 18 \\ + 0.0260 \times 17 + 0.015 \times 16 + 0.0363 \times 1 \\ = 11.59$$

CEA gives $\bar{M}_m = 11.685$

$$\frac{P_o}{P_e} = \frac{68}{1} = 68 = \left(1 + \frac{\gamma_m - 1}{2} M_e^2\right)^{\frac{\gamma_m}{\gamma_m - 1}} \rightarrow \text{we need } \underline{\gamma_m}$$

first compute $R_m = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{11.59} = 717.4 \frac{J}{\text{mol} \cdot K}$

$$\bar{c}_{p_m} = \sum X_i \bar{c}_{p_i}$$

$$\bar{c}_{p_{mon}} = \frac{\gamma_{mon} \bar{R}}{\gamma_{mon} - 1} = \bar{c}_{p_o} = \bar{c}_{p_H} = 2.493 \bar{R}$$

$$\bar{c}_{p_{di}} = \frac{\gamma_{di} \bar{R}}{\gamma_{di} - 1} = \bar{c}_{p_{o_2}} = \bar{c}_{p_{H_2}} = \bar{c}_{p_{OH}} = 4.448 \bar{R}$$

$$\bar{c}_{p_{tri}} = \frac{\gamma_{tri} \bar{R}}{\gamma_{tri} - 1} = \bar{c}_{p_{H_2O}} = 6.882 \bar{R}$$

$$\bar{c}_{p_m} = \sum X_i \bar{c}_{p_i} = 5.772 \bar{R} \quad \text{and} \quad \gamma_m = \frac{\bar{c}_{p_m}}{\bar{c}_{p_m} - \bar{R}} = 1.21$$

CEA gives $\gamma_m = 1.15$

put γ_m in (*) $\rightarrow$ Solve $\rightarrow M_e = 3.21$

CEA gives $M_e = 3.143$ (as γ differs)

$$T_o = 3429 \text{ K} \rightarrow T_e = \frac{T_o}{1 + \frac{\gamma_m - 1}{2} M_e^2} = 1647 \text{ K}$$

CEA gives $T_e = 1850 \text{ K}$

$$u_e = M_e \sqrt{\gamma R T_e} = 3.21 \sqrt{1.21 \times 7717.4 \times 1647} = 3838 \frac{\text{m}}{\text{s}}$$

$$u_{eq} = u_e + \frac{RT_e}{u_e} \left(1 - \frac{P_a}{P_e}\right) = u_e = 3838 \frac{\text{m}}{\text{s}} \quad (\text{for fully-expanded } u_e = u_{eq})$$

CEA gives $u_{eq} = 3928 \frac{\text{m}}{\text{s}}$

Note: CEA's I_{sp} is u_{eq}

$$I_{sp} = \frac{u_{eq}}{g_e} = \frac{3928}{9.81} = 391.5$$

$$\frac{A_e}{A^*} = f(M_e), \quad M_e = 3.21 \xrightarrow{\text{online calc}} \frac{A_e}{A^*} = 8.73$$

CEA gives $\frac{A_e}{A^*} = 9.0213$

consider $\text{H}_2\text{-O}_2$ reaction:

First, consider ideal reaction:

- 1) Stoichiometric reaction: $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$ (1)
- 2) Excess fuel: $(1+n) \text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O} + n \text{H}_2$ (2)
- 3) Excess O_2 : $\text{H}_2 + (\frac{1}{2} + m) \text{O}_2 \rightarrow \text{H}_2\text{O} + m \text{O}_2$ (3)

From Eq. (11.4) of the book, with assuming $u_{eq} = u_e$, which means perfect expansion), we have:

$$1) \quad I_{sp} = \frac{u_e}{g_e} = \frac{1}{g_e} \sqrt{2 \left(c_p T_o + \underbrace{\frac{\bar{Q}_R}{18}}_{M_p} \right) \left[1 - \left(\frac{P_e}{P_o} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (1)^*$$

$$2) \quad \bar{I}_{sp} = " = " \sqrt{2 \left(c_p T_o + \frac{\bar{Q}_R}{18+2n} \right) \left[1 - \left(\frac{P_e}{P_o} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (2)^*$$

$$3) \quad I_{sp} = " = " \sqrt{2 \left(c_p T_o + \frac{\bar{Q}_R}{18+32m} \right) \left[1 - \left(\frac{P_e}{P_o} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (3)^*$$

If we plot the above 3 Eq^s, we have:

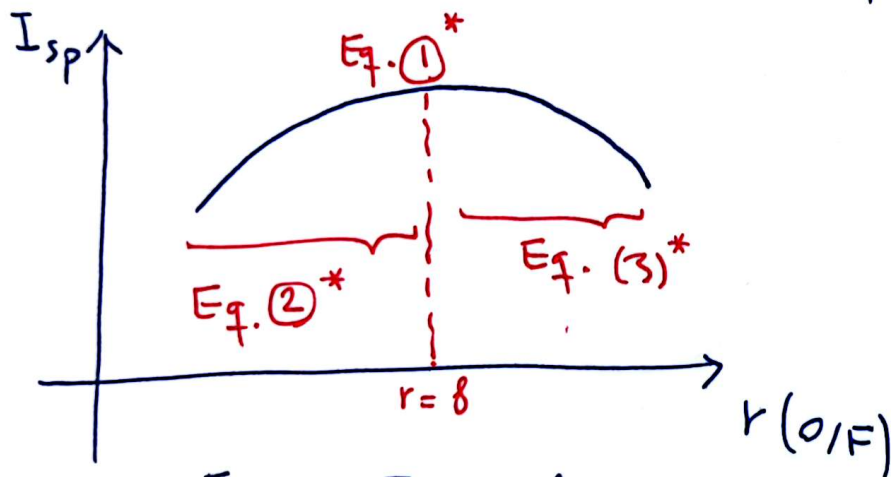


Fig 1: I_{sp} from theory

M_p is minimum (I_{sp} is maximum) if $\begin{cases} n=0 \\ m=0 \end{cases}$, i.e., stoichiometric

However, in practice, at stoichiometric conditions, product (H_2O) dissociates and M_p changes, and $\bar{Q}_R$ also changes.

So, the maximum I_{sp} in practice occurs at $r < 8$, see:

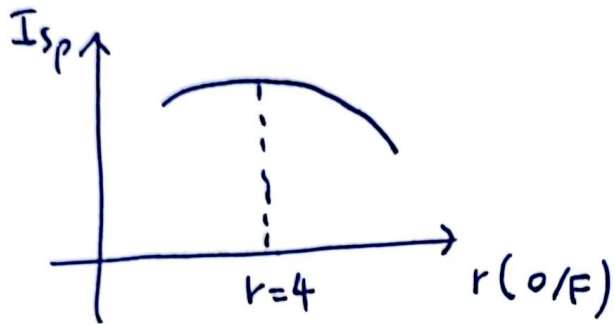
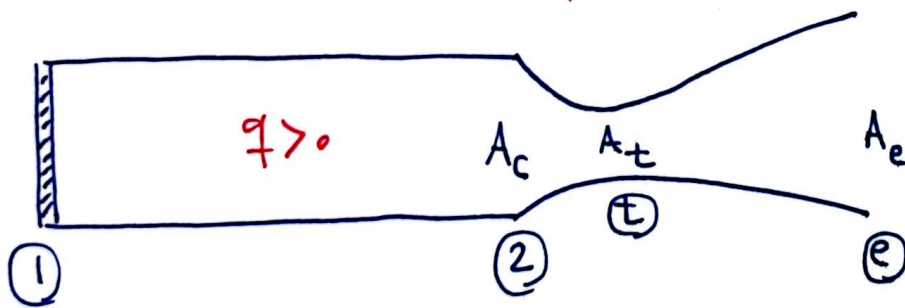


Fig 2: I_{sp} from CEA

Note: in ①*, ②*, ③* $c_p T_o + \frac{\bar{Q}_R}{M_p} = c_p T_o_2$

T_o_2 : 'temperature of products'

"Combustion chamber pressure loss"



Note: Total pressure is not constant in combustion chamber.
($P_{o1} \neq P_{o2}$)

We have:
$$\frac{P_{o2}}{P_{o1}} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \left\{ \frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2} \right\}^{\frac{\gamma}{\gamma-1}}$$
 "Rayleigh relation"

We also have:
$$\frac{P_2}{P_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2}$$

(P.31)

Usually, in combustion chamber, $M_1 \approx 0$, so Rayleigh relation simplifies to:

$$\frac{P_{0,2}}{P_{0,1}} = \frac{1}{1 + \gamma M_2^2} \left(1 + \frac{\gamma-1}{2} M_2^2\right)^{\frac{\gamma}{\gamma-1}} \quad (1)$$

$$\frac{P_2}{P_1} = \frac{1}{1 + \gamma M_2^2} \quad (2)$$

How to get M_2 ? : Exit of combustion chamber is inlet of nozzle! From $\frac{A_c}{A_t}$ and area-Mach relation, read subsonic M_2 .

$$\frac{A_c}{A_t} = \frac{1}{M_2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_2^2\right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

$\underbrace{\frac{A_c}{A_t}}_{\text{given}} \quad \rightarrow \quad \text{unknown } M_2 < 1$

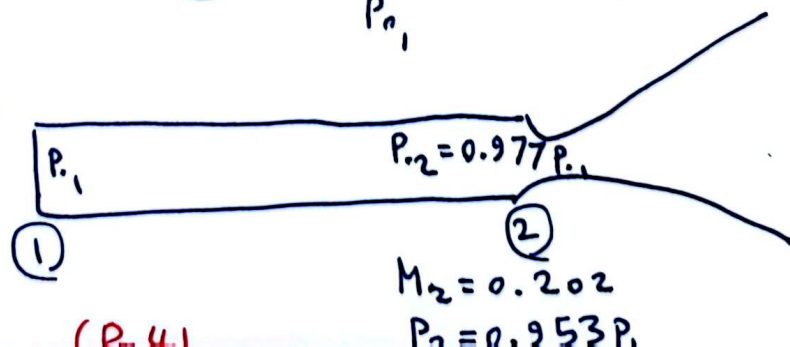
Example: We have a nozzle with $\frac{A_c}{A^*} = 3$, $\gamma = 1.2$.

Find $\frac{P_{0,2}}{P_{0,1}}$ and $\frac{P_2}{P_1}$.

Solution: First read M_2 subsonic from $\frac{A_c}{A_t} = 3$, $\gamma = 1.2$

$M_2 = 0.202 \rightarrow$ use (1) get $\frac{P_{0,2}}{P_{0,1}} = 0.977$

use (2) $\frac{P_2}{P_1} = 0.953$



Now, let's do analysis of pumps in "liquid rockets"

Define "pressure head" or head, as: $h = \frac{P}{\rho g}$

Example: $P = 1 \text{ atm} = 101325 \text{ Pa}$

1) Find water head: $h_w = \frac{101325}{1000 \times 9.81} = \boxed{10.33 \text{ m}}$

2) Find mercury " : $h_m = \frac{101325}{13600 \times 9.81} = 0.76 \text{ m} = \boxed{76 \text{ cm}}$
